Do Transformers Add Anything to a 3D CNN for Alzheimer's Disease Detection? A Leakage-Free, Statistically Controlled Benchmark on Structural MRI

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Abstract*—Deep-learning systems for Alzheimer's disease (AD) detection from MRI routinely report 95–99% accuracy, and CNN–transformer hybrids are increasingly presented as the next step. Most of these figures, however, come from slice-level splits that place images of the same patient in both training and test sets, and hybrids are rarely compared against matched single-model baselines. We ask whether a transformer branch improves a 3D CNN when the evaluation is honest. Six fully 3D models — DenseNet-121, MedicalNet ResNet-18, ViT, Swin, and two late-fusion hybrids (CNN–ViT and Swin–ViT) — were trained on 1,132 T1-weighted volumes from 395 ADNI subjects with one preprocessing pipeline, one classifier head, and identical subject-level five-fold splits, and were compared with fold-paired tests and subject-level bootstrap intervals. On CN vs. AD, the CNN–ViT hybrid reached a balanced accuracy of 0.806 (95% CI 0.757–0.852), statistically indistinguishable from DenseNet alone (0.800; paired $p = 0.54$). Replacing the CNN branch with Swin, by contrast, lowered macro-F1 in all five folds (balanced accuracy 0.670). On CN vs. MCI vs. AD, the best balanced accuracy was 0.549 against a chance level of 0.333, with MCI as the bottleneck, and a from-scratch ViT never predicted the majority class. 3D attribution maps disagreed across methods and partly highlighted non-brain tissue. At the scale of a typical ADNI study, the convolutional branch carries the signal and the transformer adds no measurable value.

***Index Terms*—Alzheimer's disease, structural MRI, 3D CNN, vision transformer, hybrid models, data leakage, benchmark, explainability**

I. Introduction

Alzheimer's disease (AD) is the leading cause of dementia, and the atrophy it causes in the hippocampus and medial temporal lobe is visible on structural MRI [1], [2]. This has made MRI-based AD classification one of the most popular applications of deep learning in medical imaging [3].

The reported performance in this field depends heavily on how the data are split (Fig. 1). Studies that classify 2D slices or curated 2D images report a median accuracy of 99.0% [4]–[11]. Studies that keep every subject entirely in either the training or the test set report 60–90% [12], [13]. The difference is largely an artefact: when Yagis *et al.* [14] kept the data and model fixed and changed only the split from slice level to subject level, accuracy fell by about 28 points. Wen *et al.* [12] found possible leakage in more than half of the CNN studies they reviewed, a recent scoping review identifies leakage as the most recurrent source of performance inflation in AD deep learning [15], and even 3D CNNs can leak through longitudinal scans of the same subject [16].

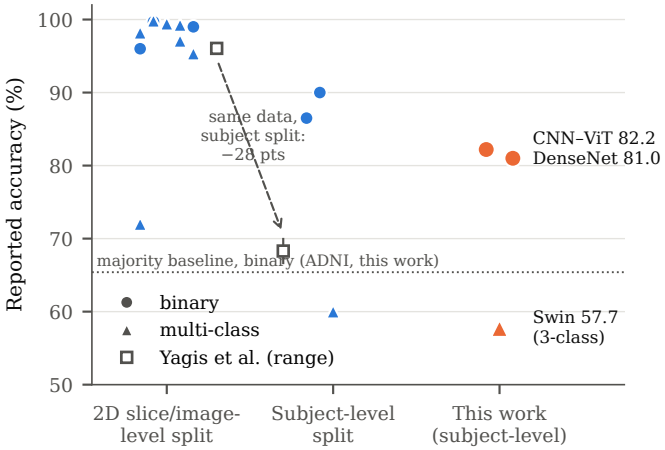


Fig. 1. **The leakage gap.** Accuracies reported in the literature, grouped by the level at which data were split. Squares show Yagis *et al.* [14], who changed only the split on the same data and model. Our subject-level results (orange) fall in the honest regime. Reported ranges are plotted at their midpoint. Dotted line: majority-class baseline of our binary task.

A second, less discussed problem concerns architecture claims. CNN–transformer hybrids are motivated by the intuition that convolutions capture local atrophy while self-attention captures long-range structure [5], [10]. Hybrids, however, are usually evaluated against weaker or differently tuned baselines, on different splits, and without statistical tests, so the contribution of the fusion itself is unknown. Under leakage-free evaluation it is not clear whether any of the gain survives.

We therefore design a controlled benchmark in which only the backbone changes, and use it to answer three questions:

- **RQ1:** Does adding a global transformer branch to a 3D CNN improve AD classification?
- **RQ2:** Which component carries the signal — the convolutional branch, the transformer, or pretraining?

- **RQ3:** Do 3D attribution maps of such models point to disease-related anatomy?

Our contributions are:

1) A leakage-free benchmark of six 3D models on ADNI with identical subject-level splits, preprocessing, classifier head, and per-task training recipe, reported with majority-class baselines, fold-paired tests, and subject-level bootstrap confidence intervals.
2) A branch ablation showing that adding a ViT to a 3D CNN gives no measurable gain ($p = 0.54$), whereas removing the CNN branch costs 0.14 macro-F1 in every fold.
3) Two documented failure modes: instability of a medically pretrained ResNet, and collapse of a from-scratch 3D ViT that never predicts the majority class.
4) A multi-method 3D attribution analysis that exposes method disagreement and non-brain attribution, together with a practical checklist for evaluating future hybrid models.

II. Related Work

Slice-level studies. ViT, ViT–GRU, multi-scale CNN, and CNN–Swin models trained on 2D slices report 95–99.8% accuracy [4]–[6], [10]. These studies use slice-level splits or curated 2D image sets, so their figures cannot be compared with subject-level results [14], [15].

Subject-level 3D studies. Wen *et al.* [12] reported 85–88% for CN vs. AD with 3D CNNs in a reproducible framework, and Bäckström *et al.* [17] showed that a subject-level split is essential for 3D CNNs. Saoud and AlMarzouqi [13] used one baseline scan per subject and an ensemble of 138 regional 3D ViTs, reporting 90% for CN vs. AD and 60% for the three-class task. A concurrent study reports 72.3% with a multi-modal 3D CNN on OASIS-1 under subject-level cross-validation [18]. None of these works isolates the contribution of a transformer branch inside a hybrid; that is the gap we address.

III. Benchmark Design

A. Leakage-Free Protocol

Let $\mathcal{D} = \{(x_i, y_i, s_i)\}_{i=1}^{N}$ be the scans, where x_i is a 3D volume, y_i its label, and s_i its subject. We partition the set of *subjects* into five stratified folds $S_1, \ldots, S_5$ (seed 42) and test on fold k after training on the remaining folds, so that

$$\{s_i : i \in \text{train}_k\} \cap \{s_i : i \in \text{test}_k\} = \varnothing. \tag{1}$$

[TODO: state where the early-stopping validation subjects come from in each fold]. Because subjects contribute 2.9 scans on average, predictions are aggregated per subject:

$$\hat{y}_s = \arg\max_c \frac{1}{|I_s|} \sum_{i \in I_s} p_\theta(c \mid x_i), \qquad I_s = \{i : s_i = s\}. \tag{2}$$

All metrics are computed at subject level. We report balanced accuracy (BAcc, the mean per-class recall), macro-F1, accuracy, and, for the binary task, sensitivity and specificity with AD as the positive class. The reference points are the majority-class baseline (always CN: accuracy 65.4% binary and 52.2% three-class) and chance BAcc (0.5 and 0.333). Accuracy and macro-F1 are reported as mean ± SD over folds. BAcc is reported with a 95% subject-level bootstrap confidence interval (10,000 resamples of the pooled out-of-fold predictions). Models are compared with a paired t-test on fold-wise macro-F1, and we also report how many folds each model wins.

B. Cohort and Preprocessing

We used ADNI [19], [20] 3T T1-weighted MPRAGE volumes (ADNI GO/2/3, pre-processed N3m NIfTI). After removing 18 off-protocol scans, the cohort contains 206 CN (627 scans), 80 MCI (182), and 109 AD (323) subjects. A deterministic MONAI [21] pipeline reorients each volume to RAS, resamples it to 1.5 mm isotropic spacing, crops the foreground, applies z-score normalization over non-zero voxels, and resizes it to 96^3 (all binary runs and the three-class ViT) or 128^3 (three-class CNN and Swin). No skull-stripping or registration is applied. Training-time augmentation uses flips, $\pm 10^\circ$ affine transforms, noise, smoothing, bias field, and gamma and intensity perturbations.

C. Controlled Model Family

All backbones produce a feature vector f that feeds a shared head (dropout 0.2 and a linear layer). We test four single backbones and two hybrids:

- **DenseNet-121** [22]: 3D, trained from scratch, 1024-D features.
- **ResNet-18** initialized with MedicalNet weights [23]: 512-D.
- **ViT** [24]: 384-D, 6 layers, 6 heads, 16^3 patches, trained from scratch.
- **Swin**: the SwinUNETR encoder [25], [26] (feature size 48, four stages).
- **CNN–ViT hybrid**: runs DenseNet and ViT on the same volume and concatenates their features, $z = [f_{\text{CNN}} \| f_{\text{ViT}}]$.
- **Swin–ViT hybrid**: the same design with Swin in place of DenseNet, so that comparing it with CNN–ViT isolates the contribution of the convolutional branch.

Fig. 2 summarizes the design.

D. Training

We use AdamW [27] (learning rate 10^{-4}) with cosine decay, linear warmup, mixed precision, an effective batch size of 8 (2 × 4 accumulation), and gradient clipping at 1.0 on one 16 GB RTX 5060 Ti.

- **Binary runs:** class-weighted cross-entropy with label smoothing 0.1 [28], weight decay 0.03, at most 100 epochs, early-stopping patience 18.
- **Three-class runs:** focal loss [29] ($\gamma = 2$), weight decay 0.01, at most 80 epochs, patience 15.

[TODO: list the imbalance corrections active in each run]. Checkpoints are selected on validation subject-level macro-F1.

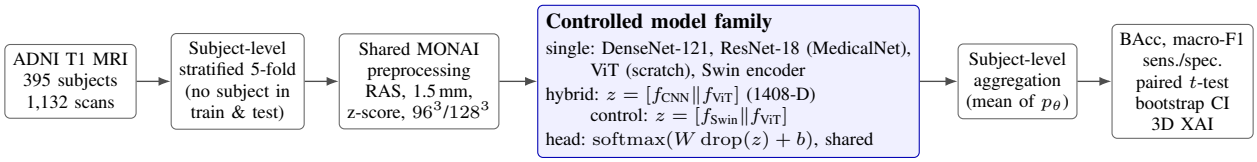


Fig. 2. **Benchmark design.** Every model sees the same subjects, splits, preprocessing, and classifier head, and models are trained with one recipe per task; only the backbone changes. The Swin–ViT hybrid is an ablation control that removes the convolutional branch.

TABLE I

CN VS. AD (315 SUBJECTS). ACCURACY AND MACRO-F1: MEAN ± SD OVER FIVE SUBJECT-LEVEL FOLDS. BACC, SENSITIVITY, AND SPECIFICITY: POOLED OUT-OF-FOLD SUBJECT PREDICTIONS, WITH A 95% BOOTSTRAP CI. Δ: FOLD-PAIRED DIFFERENCE IN MACRO-F1 AGAINST THE REFERENCE MODEL, WITH THE FOLDS WON.

Model	Accuracy	Macro-F1	BAcc [95% CI]	Sens.	Spec.	Δ macro-F1 (paired p; folds won)
CNN–ViT hybrid	**0.822**±0.057	**0.802**±0.062	**0.806** [0.757, 0.852]	0.752	**0.859**	reference
DenseNet-121	0.810±0.041	0.788±0.062	0.800 [0.752, 0.846]	**0.771**	0.830	−0.014 ($p = 0.54$; 2/5)
Swin–ViT hybrid (no CNN)	0.702±0.070	0.662±0.101	0.670 [0.615, 0.724]	0.569	0.772	−0.140 ($p = 0.11$; 0/5)
ResNet-18 (MedicalNet)	0.698±0.139	0.654±0.168	0.689 [0.635, 0.742]	0.670	0.709	−0.148 vs. CNN–ViT
Majority baseline (always CN)	*0.654*	*0.395*	*0.500*	*0.000*	*1.000*	

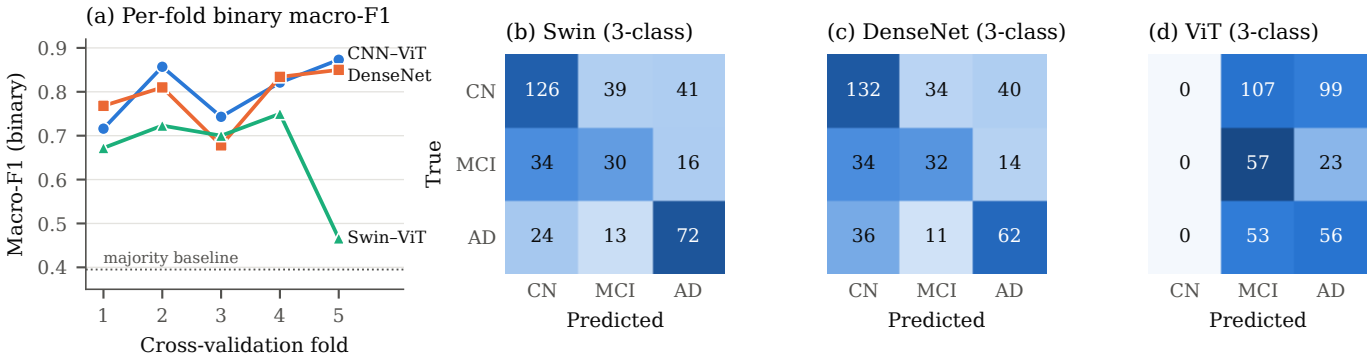


Fig. 3. **(a)** Per-fold binary macro-F1 on identical subject-level folds: CNN–ViT and DenseNet alternate, whereas Swin–ViT (no CNN branch) is lower in every fold. **(b–d)** Pooled three-class confusion matrices (subject counts; shading is row-normalized). MCI is mostly absorbed into CN and AD, and the from-scratch ViT never predicts CN.

E. Explainability Protocol

We computed Grad-CAM [30], Grad-CAM++ [31], Score-CAM, Integrated Gradients [32], SmoothGrad [33], occlusion, LIME [34], and attention rollout [35] in 3D, and inspected whether the methods agree with one another and with known AD anatomy.

IV. RESULTS

Finding 1 — Adding a ViT branch to a 3D CNN gives no measurable gain (RQ1). The CNN–ViT hybrid has the highest mean scores (Table I), but its advantage over DenseNet alone is +0.014 macro-F1. A paired t-test gives $t = 0.67$ with 4 degrees of freedom ($p = 0.54$); the hybrid wins three of five folds, and the BAcc confidence intervals of the two models almost coincide. Fig. 3a shows that the two models cross each other from fold to fold. Over all 315 subjects the hybrid classifies 259 correctly and DenseNet 255.

Finding 2 — The convolutional branch carries the signal; pretraining does not (RQ2). Removing the CNN branch (Swin–ViT) lowers macro-F1 by 0.14 and loses in all five folds. The paired t-test is not significant ($p = 0.11$) because of a single extreme fold, but the BAcc confidence intervals do not overlap (0.615–0.724 vs. 0.757–0.852), and Swin–ViT misses 43% of AD subjects. The MedicalNet-pretrained ResNet is only about four points above the majority baseline in accuracy and is bimodal across folds: three folds reach 0.78–0.82 macro-F1 and two fall to 0.449. This is consistent with a late accuracy take-off interacting with early stopping, a hypothesis we have not yet tested. At this data scale, inductive bias from convolution matters more than either global attention or medical pretraining.

Finding 3 — Three-class classification is well above chance, but MCI is the bottleneck. Swin and DenseNet are tied (paired $p = 0.52$) at a BAcc of about 0.54–0.55, well above chance (Table II). Their per-class F1 is about 0.65 for CN and 0.55–0.61 for AD, but only 0.37–0.41 for MCI. Fig. 3b–c shows that MCI subjects are mostly absorbed into CN and AD, as expected for a clinically intermediate stage. The from-scratch ViT reaches 28.6% accuracy, below the 52.2% majority baseline, because it *never* predicts CN (Fig. 3d). Its high MCI

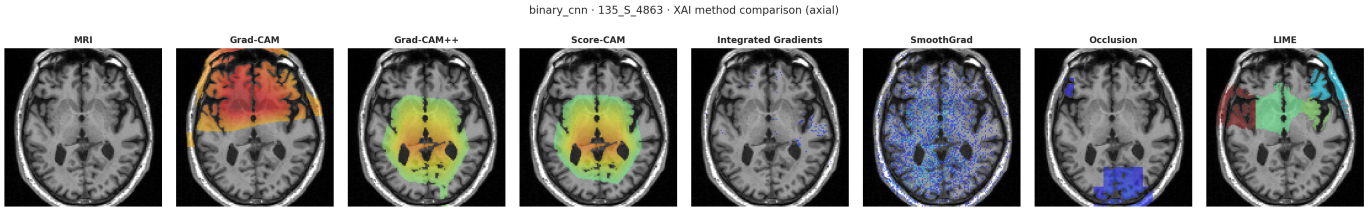


Fig. 4. **Attribution methods disagree.** Eight 3D attribution methods applied to the binary DenseNet on one AD subject (axial slice): Grad-CAM points frontally, Grad-CAM++/Score-CAM centrally, occlusion to the occipital region and skull, and LIME partly to the scalp.

TABLE II
CN VS. MCI VS. AD (395 SUBJECTS). CHANCE BACC IS 0.333.

Model	Macro-F1	BAcc [95% CI]	MCI recall
Swin	**0.537**±0.078	**0.549** [0.499, 0.601]	0.375
DenseNet-121	0.524±0.071	0.537 [0.485, 0.588]	**0.400**
ViT (scratch)	0.256±0.012	0.409 [0.363, 0.454]	0.713
Majority baseline	*0.229*	*0.333*	*0.000*

recall is therefore an artefact of predicting MCI or AD for every subject, not evidence of skill. This pattern is consistent with a data-hungry model [24] that has learned nothing useful, pushed toward the minority classes by imbalance corrections. **Finding 4 — Without skull-stripping, attribution maps are not anatomically conclusive (RQ3).** On the same subject, the attribution methods highlight different regions (Fig. 4). For the Swin and Swin–ViT models, group-averaged Integrated Gradients over 16 AD subjects is nearly uniform across thirds of the volume (0.036–0.055), with mass on the cerebellum, orbits, and scalp rather than a medial-temporal focus. Similar identity and shortcut cues have been reported for 3D MRI models [16]. A single favourable heatmap would therefore be misleading, and attribution claims require skull-stripped inputs, atlas ROIs, and sanity checks.

V. DISCUSSION

Our results fall in the same honest regime as other subject-level studies (Fig. 1): 82% binary accuracy, against 85–88% [12] and 90% [13], and 58% three-class accuracy against 60% [13]. We obtain these with a single model and without registration or skull-stripping, which plausibly explains the remaining gap. The central lesson concerns architecture claims. At around 300–400 subjects, the benefit usually attributed to hybrids came from the convolutional branch, and it disappears when a hybrid is compared with an equally tuned CNN on the same splits.

Checklist for evaluating hybrid models. Based on these findings, we recommend:

1) Split at the subject level and aggregate scans per subject.
2) Report the majority-class and chance baselines next to every score.
3) Compare a hybrid against each of its branches trained alone, on identical folds.
4) Test differences with fold-paired tests and confidence intervals, and report the number of folds won.
5) Report per-fold results, not only mean ± SD.
6) Skull-strip the inputs and use more than one attribution method before making anatomical claims.

Limitations.

- *Incomplete model grid.* Standalone ViT and Swin were not run on the binary task, and CNN–ViT was not run on the three-class task.
- *Statistical power.* Five folds and a single seed limit the power of the tests, so a small benefit of hybridization cannot be ruled out.
- *Possible age confound.* AD subjects are older on average than CN subjects, and no age-only baseline was run.
- *External validation.* The study uses only the ADNI cohort.

VI. CONCLUSION

Under leakage-free, statistically controlled evaluation, a transformer branch did not improve a strong 3D CNN for AD classification. The convolutional branch carried the signal, the three-class task was limited by MCI, and attribution maps on non-skull-stripped inputs were not anatomically conclusive. Hybrid-model claims should be backed by matched single-branch baselines and paired tests. Future work will complete the model grid with repeated cross-validation, add skull-stripping, atlas-ROI attribution, and an age-only baseline, and validate the models on independent cohorts.

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